BreakBite · Lab 3 final report

PRM393 · Cloud-Enabled Mobile App (Firebase, ARTEMIS, AI code review) · Team BreakBite (5 members) · repository https://github.com/baonv0817/PRM393_Lab03_Group2 · pinned versions and the map of deliverables: README.md.

> **How to read this report.** Every claim that something works points to a file in the repository (a test, a screenshot in `evidence/` or a document in `docs/`). What could not be shown yet is listed in section 11 instead of being filled in. Screens of P1 come from a real Android phone with the real Firebase. Screens of P2 were captured on an Android emulator, first with fake data and then again with the real Firebase (`evidence/p2-real/`). The FCM, Analytics and ARTEMIS evidence comes from the real Firebase project `breakbite-1315b`.

1. Project overview and Lab 2 recap

BreakBite is a pre-order, pick-up-at-the-counter food app for students who study on Campus District 9 and have a short break between classes. A **customer app** (S01–S30) lets them order, pay (simulated), follow, edit and review an order; a **seller app** (K01–K15) lets kiosk staff take orders, run the menu and see their shift. Both are one Flutter project; the account role chooses the app. The design is the Lab 2 Figma (`PRM323-Lab2-Group2`, frames with the v2 suffix).

Lab 2 success criterion: finish an order in under 35 seconds with at most 8 taps (docs/analytics-plan.md; the `purchase` event is the one that measures it). **Flows:** F1 quick order, F2 order with a voucher, F3 payment error then cash; plus sign-in, search, export and upload, push notification and the seller side. Main screens built: **44 of 44** (K07 is not in Figma), no placeholder route is left (checked in `lib/routing/routes_p*.dart`).

Member	Area
P1	Ordering flow S03–S08, S21, S22, S25, S29; architecture doc, design traceability, shared theme and widgets
P2	After ordering S09–S12, S20, S23, S24; seller menu K08–K11, K14, K15; PDF receipt export
P3	Firebase: Auth, Firestore, rules, Cloudinary, Crashlytics, Remote Config; S15–S17, S26, S27, Firebase demo area
P4	Seller order handling K01–K06, K12, K13; code review; report, slides, video
P5	S13, S14, S18, S19, S28, S30; FCM; analytics; ARTEMIS tests

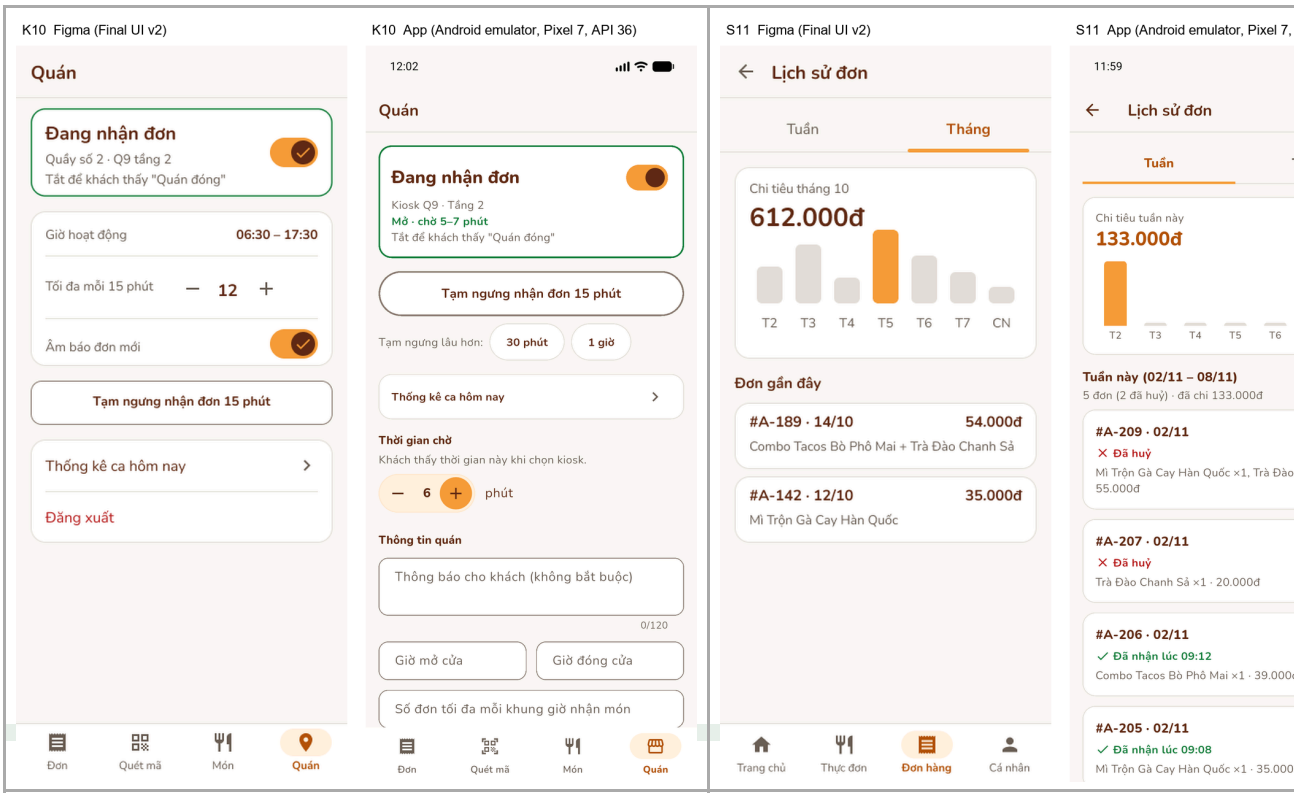
2. Design traceability and deviations

docs/design-traceability.md maps every Lab 2 screen, flow and component to a route, a widget and a data source, with a status. Section 5 of that file records **every place where the app differs from Figma, with the reason** (D01–D33). The main ones:

- **D01, D03:** Google Sign-In replaces the school e-mail + OTP, and a signed-out user only sees S26 (handout requirement).
- **D12, D13:** the add-to-cart stepper and the floating cart button replace the + icon and the cart icon of S02 (the cart must be reachable and correctable).
- **D18:** S25 offers the kiosk's break slots (with a capacity) instead of a typed timetable, because the data model books slots.

- **D23, D24:** K14/K15 have no photo box and no "Xoá món": the final assignment drops Cloud Storage and the rules forbid deleting menu items.
- **D21, D22, D25–D27:** the screens of P2 add what the data model already had (wait time, notice, hours and capacity in K10; the line "Đang xử lý" in K11; weekly and monthly groups in S11; a timeline, review and receipt in S12).

The column "Figma updated" is still **Not yet** for these rows; updating Figma is task T83 (section 11). Two examples of Figma (left) and the app (right) from [evidence/figma/]():



K10 "Quán" (D21) and S11 "Lịch sử đơn" (D25). Fourteen such pairs for the screens of P2 are in `evidence/figma/`, thirteen more for P1.

3. Architecture and MVVM

MVVM with Provider and go_router, following the official Flutter architecture guide: a **UI layer** (screens and ViewModels grouped by feature, `lib/ui/<feature>/`) above a **data layer** (repositories and services, `lib/data/`) above pure models (`lib/domain/`). Dependencies only point down. [docs/architecture.md]() explains why the folders differ from the handout's suggestion (five people working in parallel; the final assignment needs a shared package).

Layer	Responsibility	Knows about
Screen `ui/*/widgets`	Draws the ViewModel's `ViewState`, forwards taps	Its ViewModel, `ui/core`, routes
ViewModel `ui/*/view_models`	Screen state, validation, calls to repositories and services, `Command` for actions	Interfaces and models only
Repository `data/repositories`	Business data in Firestore → models, queries, transactions	Models, Firestore (in its implementation)
Service `data/services`	Wraps one SDK (Auth, FCM, Analytics, Crashlytics, Remote Config, Cloudinary)	The SDK (in its implementation)

Model `domain/models`	Immutable data and pure rules	Nothing

Enforced by a test: `test/architecture_test.dart` fails if a screen, a ViewModel, a route or a model imports an SDK, if a screen holds a `Color`, `Colors` or `TextStyle` literal, or if a ViewModel does not use `SafeChangeNotifier`. Fake data lives in `dev/`, outside `lib/`, so the submitted build has no mock data. **MVVM check across the repo (T72):** every widget file was read against the rules; three pieces of logic were moved out of `build()` into ViewModels (S11, S12, K11) and two cases were kept with a reason (section 3 of the architecture document).

4. Design system in code

Tokens live in one place, `lib/ui/core/theme/`: `app_colors.dart` (colors from the Figma v2 frames, with contrast tests for WCAG), `app_dimens.dart` (spacing, radius, minimum tap target 48 dp) and `app_theme.dart` (the Material theme built from them). Components are one widget each in `lib/ui/core/widgets/` (`AppButton`, `AppChip`, `AppTopBar`, `QuantityStepper`, `BottomActionButton`, `ViewError`/`ViewEmpty`/`LoadingView`, `StatusBadge`…); every screen uses them, and the architecture test forbids colour or text-style literals in screens. The widget catalog `dev/widget_catalog.dart` shows every component and variant and is tested at 360 and 412 dp with text at 1.0 and 1.3: no overflow, every tap target at least 48 dp, every one labelled. Every state a screen can be in (loading, empty, error with retry) is a shared widget, not a copy.

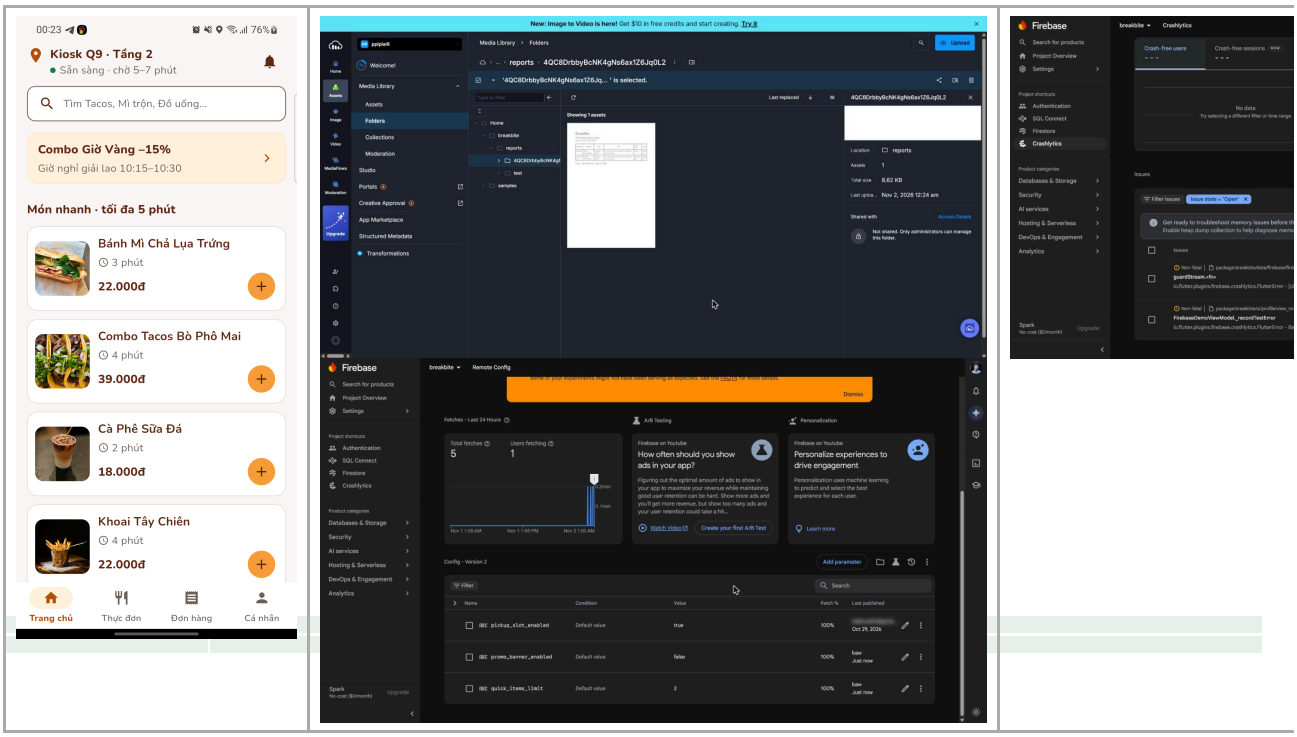
5. Firebase design

Project `breakbite-1315b`, Spark plan. Screens never call an SDK: each service is an interface in `lib/data/services/` with a Firebase implementation, and every failure comes back as a typed `AppException` with a Vietnamese message (docs/firebase-design.md has the full table per service).

Service	What the app does	Failure handling
Authentication	Google Sign-In (S26); first sign-in creates `users/{uid}` with role `user`; the role (set with `set-role.mjs`) routes to the customer or the seller app; sign-out clears local data	Cancelled sign-in, no network and "no token" each show their own message; the router waits on a splash screen until the sign-in state is known
Firestore + rules	Menu, cart, orders (placed, edited and cancelled in one transaction), incidents, reviews, notifications; deny by default, access by role, staff only at their own kiosk; 88 rules tests and an integration test against the real rules	A write that depends on current data runs in a transaction, so the user sees a clear reason (for example "Đơn này vừa được xử lý ở nơi khác") instead of a rules error
Files (Cloudinary)	The order receipt (S12) and the report in the demo area are PDFs uploaded with a progress bar to `breakbite/reports/{uid}/`; the URL and metadata are kept in `reports` (owner-only rules)	Not configured, larger than 5 MB, no network, timeout and "PDF cannot be made" each have a message; the retry button stays
Cloud Messaging	Permission on Android 13+, token per user in `users/{uid}.fcmTokens`, messages saved to the in-app list (S19) in foreground, background and terminated states; tapping opens the order	A refused permission or a failed token write never blocks sign-in or sign-out
Analytics	11 events in the plan, all logged in code; `setUserId` after sign-in, no personal data	Logging never throws

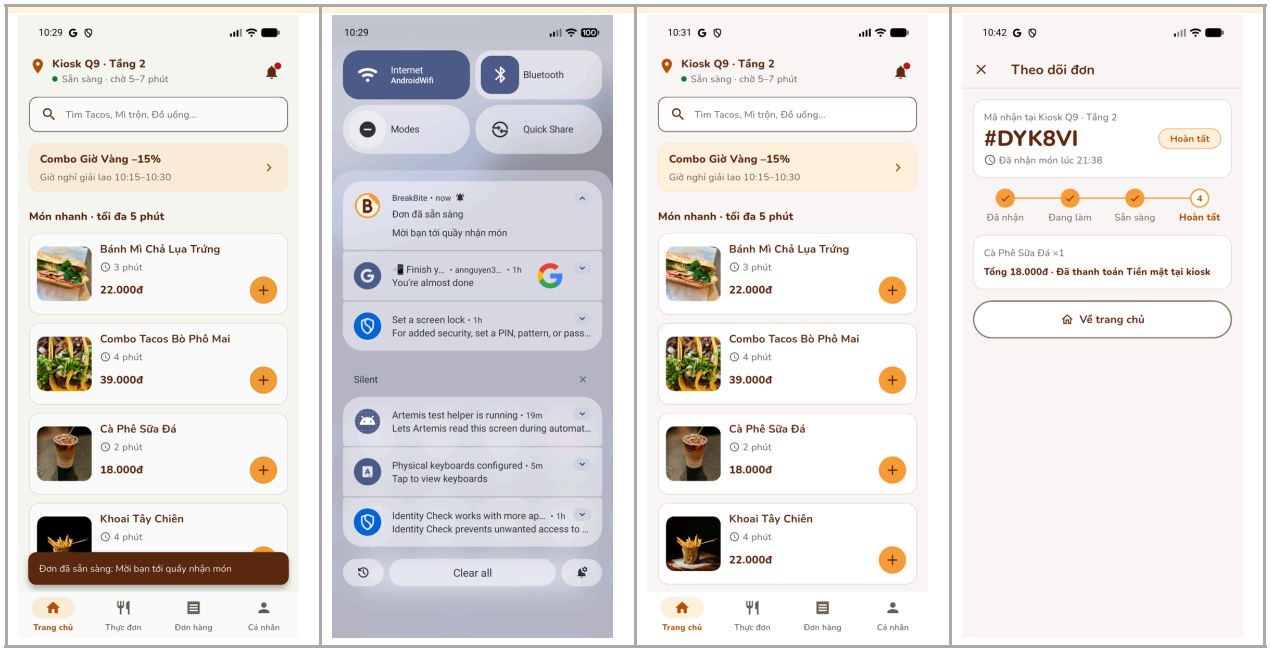
Crashlytics	Uncaught errors reported; handled errors from real error paths recorded as non-fatal with the keys `role` and `kiosk_id`	Reporting never throws
Remote Config	`quick_items_limit`, `promo_banner_enabled`, `pickup_slot_enabled`; the app listens for updates, so Home changes without a rebuild	Offline: last values, or the in-app defaults on a first launch

Why Cloudinary: Cloud Storage for Firebase needs the Blaze plan (a payment card). The upload goes through the same `StorageService` interface with an unsigned preset, so the app holds no API secret; this is a deviation from the handout (listed in the README). The lecturer confirmed in writing that Cloudinary or S3 is acceptable.



Signed-in app (`evidence/auth`), the uploaded PDF in the Cloudinary library (`evidence/storage`), the Remote Config console (`evidence/remote-config`) and a non-fatal error in Crashlytics (`evidence/crashlytics`), all from project `breakbite-1315b`.

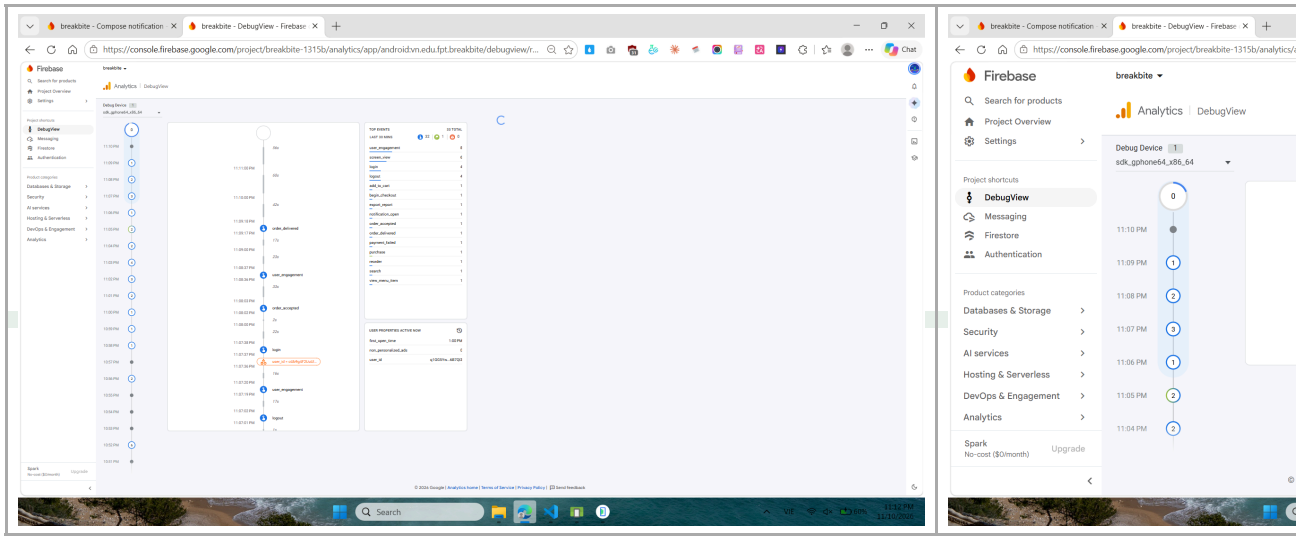
Push notifications in the three app states. A test message from the Firebase console, sent to the device token with the data `orderId`, shows an in-app banner when the app is open, a system notification when it is in the background and again when it is closed. Tapping a notification opens the tracking screen of that order, and each message is saved in the notification list (S19). Evidence: `evidence/fcm/`. **Bug found by this test:** from a closed app, the tap opened the order for a moment and Home replaced it when the splash screen redirected. `PushRouter` now waits until the app is ready (`lib/app.dart`). The screenshots before and after the fix are kept.



Open app (banner) · background (notification) · closed app, tap before the fix (ends on Home) · after the fix (opens the order).

6. Analytics plan and evidence

docs/analytics-plan.md was written before the code: 11 events covering sign-in (`login`), sign-out (`logout`), search, opening an item (`view_menu_item`), ordering (`add_to_cart`, `reorder`, `begin_checkout`, `purchase`, `payment_failed`), the seller steps (`order_accepted`, `order_delivered`) and the export (`export_report`, with `source: order_detail` for the receipt of S12). Each event answers a product question and maps to a Lab 2 step; the one that tests the success criterion is `purchase` (≤ 8 taps, < 35 seconds). Names come from `AnalyticsEvents`, parameters are numbers and short strings, never a name or an e-mail. All 11 are logged in code and covered by ViewModel tests. DebugView (debug mode on with `adb shell setprop debug.firebase.analytics.app vn.edu.fpt.breakbite`) shows the events arriving with their parameters.



DebugView: the events of the plan with their counts · the timeline of one session.

7. ARTEMIS test plan, results and exploratory session

docs/test-plan.md has 31 test cases (TC-01 to TC-31) covering sign-in and sign-out, F1–F3,

search, export and upload, push, the seller flow, the seller menu, a sold-out item and a closed kiosk, with happy paths, error paths and native interactions (the Google account chooser, a notification). e2e_test/ has one pytest function per case; each sends one instruction and **asserts on the value the agent reports back**, not only on `succeeded`; `run_repeat.py` runs a case N times, restarts the app before each run and never overwrites an earlier run. P2's cases are TC-18 to TC-29 (`test_order_manage.py`, `test_seller_menu_shop.py`).

Results. The graded suite is the 11 cases that cover every item of section 7 of the handout: TC-01 sign-in through the native Google account chooser, TC-02 sign-out, every Lab 2 flow end to end with its error branch (TC-03 F1, TC-04 F2, TC-05 F3, TC-12 seller), TC-06 search with no result (error path), TC-07 every tab, TC-08 each kind of detail screen, TC-09 export and upload and TC-10 a Remote Config value on screen; TC-03 also checks that its Analytics `purchase` event was logged (Logcat). Each ran at least 3 times on an Android emulator (Pixel 7, API 36) against the real Firebase project with the Flash profile (Gemini 3.5 Flash Lite): runs 1 to 5 on 11/11, runs 6 to 39 on 12/11. Every run is a JUnit file in `evidence/artemis/run-*.xml`, and section 3 of the test plan gives the trace id of each. TC-11 and TC-18 to TC-31 are written but kept as a backlog (not run); TC-14 to TC-17 were extra cases in the first session.

Case	Runs	Pass	Earlier failures and their fix
TC-01 sign-in (native), TC-02 sign-out, TC-12 seller accepts and delivers	3 each	3/3	None
TC-06 search with no result, TC-07 main tabs, TC-08 item and order detail, TC-09 export and upload	3 each	3/3	None (11/11)
TC-03 F1 quick order + Analytics	3	3/3	0/3 before: the agent reported the order code as the pickup code (instruction now names both)
TC-04 F2 wrong voucher	3	3/3	0/3, 0/3, 2/3 before: ARTEMIS typing (local patch), the cart's two red lines, a missed error line (instructions rewritten); one run hit the daily model quota
TC-05 F3 payment error	4	3/4	**App bug found and fixed** (below); the one failure on the fixed build was an ANR while the host also ran `flutter test`
TC-10 Remote Config quick items	3	3/3	2/3 twice before: "scroll down" made the agent swipe the wrong way (instruction now says swipe up and list the names)
Extra, 11/11: TC-16 default payment, TC-17 stepper, TC-14 change kiosk, TC-15 student profile	3 to 4	3/4, 2/3, 1/3, 1/3	Agent misreads; by hand the app was right (test plan 3.2)

Analysis. Each failure was traced to its cause in the agent's trace. The tool: this ARTEMIS version types by pasting, which Flutter fields ignore, so BB20 never reached the field; a 12-line local patch sends ASCII with `adb input text` (e2e_test/artemis-ascii-input.patch). Ambiguous instructions (TC-03, 04, 10): rewritten. Timing: one ANR under host load. And one **app bug**: after MoMo was declined and cash was chosen, the order was created and the cart emptied, but the screen stayed on "Thanh toán chưa thành công" in 3 of 4 attempts. `CheckoutScreen` navigated with its own `context` after awaiting `placeOrder`; it and `GatewayScreen` now read the router before the await, every passing run on the fixed build opened the order, and `flutter test` still passes.

The runs also showed two low usability problems (an active order card that cannot be tapped; "Mã nhận tại …" printed above the order code), recorded in section 3.3 of the test plan. The pinned ARTEMIS returns no structured output, so `e2e_test/conftest.py` reads the agent's final JSON report from the server trace.

Other checks that were run: 752 unit, ViewModel, widget and full-app flow tests (`flutter test`, no device), 88 Firestore rules tests (`firebase/tests`) and 15 integration tests of the real Firebase code against the Auth and Firestore emulators (`integration_test/firebase_emulator_test.dart`). `flutter analyze` has no issue.

Exploratory session (ARTEMIS Pro profile)

12/11/2026, **15.8 minutes**, ARTEMIS **Pro** profile on the emulator with the real Firebase project and the fixed build, started by `e2e_test/exploratory.py`: two unscripted charters back to back. Charter 1 (6.0 min, 20 steps): menu and categories, item options and a kitchen note full of special characters, cart and vouchers, search, switching kiosk and back, the system Back key. Charter 2 (9.7 min, 37 steps): orders (active, history, detail, enlarged QR, receipt export), profile (student profile with a wrong then a right phone number, default payment method, Firebase demo area, notification settings). Evidence: `evidence/artemis/exploratory-2026-11-12.json` (every step and the agent's notes) and the two screen recordings (docs/test-plan.md section 4.1). The agent left its "Issues" note empty, so the issues were read from its step log and each was repeated by hand:

Found	Severity	Handled
Order detail (S12) offers "Xuất biên nhận PDF" for a cancelled, unpaid order (#4DZ25H); the receipt is generated (9 KB) and listed under "Tệp của tôi", although the order was never paid	Medium	Open, to decide with P2 (hide the button or print "ĐÃ HUỶ")

Not problems once checked by hand: a spinner on notification settings (it loads in under a second), a "Xem giỏ" tap the agent aimed at the keyboard area (the keyboard closes and the action is reachable), a phone number the agent's typing did not reach. Seen working: validation messages, the kiosk-switch warning, vouchers, the enlarged QR, order history, the default payment method, the Remote Config fetch, export and upload.

An earlier manual session (10/11, emulator with fake data, text at 200 %, fast taps) found five low-severity layout and wording problems; the K10 one is fixed (test plan section 4.2).

8. AI code review findings and decisions

Tool: Claude Code, skill `/code-review`, run on the repositories and the screens of P2; the unchanged output is in evidence/code-review/claude-code-review.md and the full table in docs/code-review.md. Each finding was opened in the code before it was decided, because the tool can be wrong (for CR-10 its text said the description is untrimmed, but the repository trims it).

CR-01 · security · `firestore_order_repository.dart:52` · fix: —

Finding: The order total is built from client prices, never read from `menu_items`.

Decision: Accepted, not fixed: needs menu reads in the order transaction and a rules change on the ordering path of P1/P3; recorded as a known limitation.

CR-02 · bug · `firestore_menu_repository.dart:91` · fix: d6d8113

Finding: A new item with an existing name overwrites that item.

Decision: Accepted: silent data loss between two staff members.

CR-03 · bug · `firestore_incident_repository.dart:98` · fix: d1291a6

Finding: Take/answer an incident checked a stale copy.

Decision: Accepted: rules let staff update a resolved incident, so a second answer overwrote the first; now a transaction.

CR-04 · performance · `firestore_notification_repository.dart:55` · fix: 9243bf3

Finding: "Mark all read" in one write batch.

Decision: Accepted: Firestore caps a batch at 500 writes; now batches of 400, tested with 450.

CR-05 · bug · `order_history_view_model.dart:98` · fix: 2980559

Finding: "Xem đơn cũ hơn" could add nothing while older orders exist.

Decision: Accepted: the first load already skips such pages; test fails before, passes after.

CR-06 · bug · `seller_menu_view_model.dart:41` · fix: ba8dc33

Finding: First "Có lại lúc" option ignored seconds.

Decision: Accepted: 10:15:20 offered 10:30, 20 s under the promised 15 minutes.

CR-07 · bug · `firestore_order_repository.dart:275` · fix: —

Finding: Slot release fails if the slot document is missing.

Decision: Rejected: the rules allow no slot delete and keep `booked` ≥ 0; no script deletes slots.

CR-08 · maintainability · `shop_view_model.dart:134` · fix: —

Finding: `_apply` returns ok when busy.

Decision: Rejected: the screen disables every control while a write runs, and a test covers the guard.

CR-09 · performance · `shift_stats_view_model.dart:116` · fix: —

Finding: K11 reads all kiosks to get one opening time.

Decision: Rejected: 2 kiosks, 2 reads per refresh; a new repository method costs more than it saves.

CR-10 · test gap · `incident_view_model.dart:128` · fix: —

Finding: Submit does not check status or an earlier report.

Decision: Rejected: partly wrong; the button is disabled while running and the rules refuse a second report.

Five fixes, each in its own commit that names the finding; the new tests fail on the old code and pass on the new (CR-05 was run both ways). CR-02, CR-03 and CR-04 are also covered by the emulator integration test.

9. Challenges encountered

- **Cloud Storage is not free.** It needs the Blaze plan, so files go through Cloudinary behind the same `StorageService` interface; the lecturer agreed in writing.
- **Five people on 44 screens.** One folder per feature and one route file per owner kept

merges small; a few shared files (theme, widgets, router) needed rules in `CONTRIBUTING.md`.

- **Rules and UI must agree.** A check done only in the UI is not a security rule: the incident rules let staff update a resolved report, and an order's prices are not checked against the menu (CR-01, CR-03). Testing the real rules on the emulator found both kinds of gap that fake data hides.
- **Stale data on screen.** Two staff members can hold different copies of the same document; the fixes read the document again inside a transaction.
- **Evidence needs accounts.** Real-Firebase screenshots need a staff role (set with `set-role.mjs`) and the SHA-1 key of every machine. The screens were first captured with fake data and then repeated on the real Firebase.
- **The real Firebase found what the emulator hid.** The seller shift screen (K11) failed because its composite indexes were missing (16 indexes are now in `firestore.indexes.json`), and a push tapped on a closed app ended on Home (fixed in `lib/app.dart`).
- **An AI test agent is a flaky tester.** ARTEMIS used 20 to 50 model requests per test, so the free daily quota (500 requests) ran out after about 25 sessions. It mistyped, miscounted and missed snackbars that vanish in seconds. Each failure was repeated by hand or read in the trace to decide whether the app, the tool or the instruction was wrong.
- **The test device can be the problem.** Google sign-in on the emulator failed with "Account reauth failed" until its automatic Private DNS was turned off.

10. Lessons learned

- Write the traceability and the test plan **before** the screens: every deviation found later cost a table row and a Figma change.
- Put logic in ViewModels from the start: moving it out of `build()` afterwards (T72) was three small changes only because the architecture test already blocked the worst cases.
- Test against the real rules early. The emulator test caught assumptions that the fakes could not (for example that a customer may report an order that is not delivered yet: the rules allow it, the UI does not offer it).
- An AI review is a list of questions, not of facts: four of ten findings were rejected after reading the code and the rules, and one claim in an accepted-looking finding was wrong.
- Judge a failing agent test by reading its trace and repeating the step by hand, and write into the test what the agent can read reliably (a saved value, not a snackbar that disappears). Of the failures on 12/11, three came from our own instructions, one from the tool, and one was a real app bug that manual testing had missed.

11. Remaining work

Item	Note
Open problems found by ARTEMIS: a receipt can be exported for a cancelled order (medium); an active order card cannot be tapped; "Mã nhận tại …" above the order code (low)	Test plan sections 3.3 and 4.1; to decide with P1 and P2
Figma updated for the deviations D01 to D33	Listed as "Not yet" in the traceability table

Known limitation CR-01: order prices are not checked against the menu on the server	Needs a rules change on the ordering path

12. AI tools used

Tool	Purpose
Claude Code (Claude Sonnet 5.5, Claude Opus 5.5)	Writing and testing the code of P2, the ARTEMIS cases TC-18 to TC-29, the evidence screenshots, the ARTEMIS runs and their analysis, the checkout fix, the review fixes, this report (`scripts/build_report.py`) and the slides (`scripts/build_slides.js`, pptxgenjs)
Google ARTEMIS (Flash and Pro profiles) with Gemini 3.5 Flash Lite	The UI test agent and the exploratory session of section 7
Microsoft Edge TTS (vi-VN-HoaiMyNeural)	The narration of the demo video
Claude Code, skill `/code-review`	The AI code review of section 8
Other members	To be added by each member in the README and in docs/code-review.md